CDCCNet: Cross-Domain Consistency-Constrained Domain Generalization Network for Remote Sensing Optical Image Change Detection

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***Abstract*—Remote sensing change detection (RSCD) aims to identify and localize changes in the same geographical region using bi-temporal or multitemporal images. However, significant feature distribution shifts commonly exist not only between training and real-world data but also between paired images acquired at different times within the same dataset, severely limiting model generalization. Domain-generalized RSCD seeks to learn domain-invariant representations from source domains (seen), enabling direct deployment to target domains (unseen) without requiring target-domain data during training. Existing methods typically attribute performance degradation to style discrepancies and attempt to suppress style variations through feature regularization. However, because style and content information are highly coupled, such strategies often discard critical content representations. In addition, style discrepancies across datasets can further introduce change-domain shifts, which are largely overlooked by existing methods. To address these issues, this article proposes a domain generalization network for RSCD that enables models trained solely on a source domain to generalize effectively to target domains. Specifically, a feature constraint (FC) mechanism is introduced at the encoder stage to mitigate style interference through covariance alignment while preserving essential content information. At the decoder stage, a cross-domain learning (CDL) module is designed to construct a more discriminative embedding space and separate features prone to misclassification. Extensive experiments demonstrate that the proposed method achieves strong robustness and superior detection accuracy across both source and target domains, significantly outperforming existing methods.**

***Index Terms*—Change detection, contrastive learning, domain generalization (DG), domain shift, feature constraint (FC).**

I. Introduction

REMOTE sensing change detection (RSCD) is one of the important topics in remote sensing applications, aiming to accurately identify changes in targets within images of the same geographic region acquired at different times. This task plays a critical role in urban monitoring, disaster assessment, and environmental protection [1].

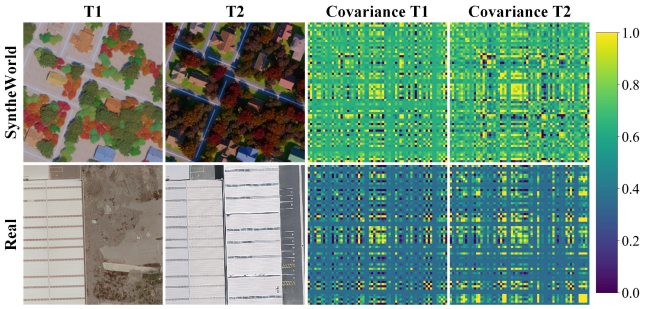


Fig. 1. Comparison of feature statistics (covariance) between multitemporal images in synthetic and real RSCD datasets. The feature covariance matrix characterizes the style of an image. It can be observed that even within the same dataset, there exist significant differences between images from different temporal phases.

In recent years, deep learning (DL) has significantly advanced the development of RSCD due to its remarkable performance across various tasks [2]. These methods typically rely on large-scale annotated datasets; however, acquiring such annotations is highly challenging. In particular, RSCD requires paired bi-temporal data for training, which further increases the difficulty of constructing large-scale datasets. To mitigate this issue, synthetic data are widely used to compensate for the shortage of training samples. However, synthetic data often exhibit substantial distributional discrepancies compared with real-world data, resulting in suboptimal performance in practical applications. Moreover, feature distribution discrepancies exist not only between bi-temporal image pairs within the same dataset but also extensively across different datasets, as illustrated in Fig. 1. Consequently, when models are deployed across domains or applied to unseen datasets, their performance may degrade severely or even fail completely.

To address this challenge, domain adaptive change detection has been proposed. Specifically, these methods aim to reduce the discrepancy between the source and target domains through distribution alignment techniques [3], [4]. However, they require target-domain samples during training, which limits their applicability in practical cross-domain scenarios where the target domain is unavailable in advance.

Meanwhile, domain generalization (DG) has emerged as a promising approach for mitigating domain shift. The

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core objective of DG is to train models exclusively on source-domain data while learning domain-invariant representations, thereby enabling direct generalization to unseen target domains. DG has been extensively studied in related areas like image segmentation and image classification. Based on different technical paradigms, existing DG techniques are generally divided into domain randomization (DR) and feature normalization (FN). DR enhances training data by generating variations in images, which can be achieved through style transformation [5], [6] or modification of feature representations [7], [8]. Exposing models to a broader spectrum of visual domains helps DR mitigate overfitting to specific training-style patterns. However, the strong dependence of DR on auxiliary domains limits its practical applicability. In contrast, FN eliminates domain-specific style characteristics by leveraging feature statistics—such as normalization [9] or whitening transformations [10], [11]—thereby mitigating overfitting to the source domain. Nevertheless, while FN can effectively remove style-related information, the strong coupling between style and content often leads to the loss of essential content information during de-stylization. When directly applied to change detection tasks, this issue may further degrade model performance. These limitations make them difficult to be directly and effectively applied in RSCD.

Currently, research on DG in the field of RSCD remains relatively limited. Specifically, GenerChange [12] employs evidential deep learning to mitigate epistemic uncertainty caused by the absence of target-domain imagery. Based on causal disentanglement theory, CDDGNet [13] achieves sufficient decoupling between change-related features and domain-specific features through frequency-domain processing and a feature reattention mechanism. However, these methods overlook a critical issue: the image style discrepancies among change detection datasets from different domains can further induce shifts in the change domain. Such domain shifts tend to cause the model to confuse genuine change features with pseudo-change features, thereby degrading its generalization capability.

To address the aforementioned issues, this article proposes a novel domain-generalized change detection model for remote sensing imagery, termed the cross-domain consistency-constraint network (CDCCNet). Specifically, the proposed model integrates feature constraint (FC) and cross-domain learning (CDL) mechanisms to effectively alleviate the above challenges. Inspired by the domain adaptation paradigm, this article introduces color and texture perturbations to source-domain image pairs to generate images with significant style discrepancies while preserving consistent semantic content, thereby constructing a pseudo target domain during training. This process provides the foundation for learning domain-invariant representations across multiple domains. Subsequently, two weight-sharing baseline networks are employed to perform feature extraction and change prediction on the original image pairs and the augmented image pairs, respectively. During the encoder stage, an FC mechanism is introduced to enforce consistent encoder responses under different styles, thereby preserving critical content information rather than simply removing style-related components. During the decoder stage, a CDL module is utilized to guide the decoder in constructing a discriminative category embedding space and effectively disentangling easily confused feature representations, thereby enhancing the model's capability to distinguish different categories. Extensive experimental results demonstrate the effectiveness of the proposed mechanism for domain-generalized change detection. The main contributions of this article are summarized as follows.

1) A novel CDCCNet is proposed for domain-generalized change detection, enabling models trained solely on source-domain data to generalize directly to previously unseen target domains. In addition, the proposed framework can further improve the performance of existing models under single-domain scenarios.
2) An FC module is designed, consisting of an auto-covariance matching loss (ACML) and a cross-covariance diagonal loss (CCDL). Specifically, ACML is designed to alleviate the influence of style discrepancies, whereas CCDL focuses on preserving content-related information. Through their joint optimization, the proposed module enables the model to learn feature representations that are robust to style variations.
3) We propose a CDL module, including spatio-temporal consistency contrastive learning (STCCL) and pseudo-change rectification contrastive learning (PCRCL), trained using masks. Specifically, STCCL generates a discriminative embedding space, whereas PCRCL decouples easily confused similar-class features, thereby enhancing model robustness.

II. Related Work

A. Change Detection

In remote sensing, change detection focuses on identifying target variations across images captured at different times. Recently, driven by advances in deep learning, numerous DL-based approaches for RSCD have been developed and demonstrated improved performance. Fully supervised methods usually depend on large annotated datasets. For example, Daudt et al. [1] proposed three fully convolutional neural network architectures to train RSCD models from scratch in an end-to-end fashion. Chen et al. [2] were among the first to apply attention mechanisms for modeling spatiotemporal relationships in RSCD. This method extracts features from paired images using a shared-weight ResNet18 and employs a local Transformer encoder–decoder module to model spatiotemporal correlations, thereby achieving high accuracy. However, the superior performance of these methods is built upon the assumption that the training and testing samples are independently and identically distributed. Once deployed in real-world scenarios with significant domain discrepancies, their performance often deteriorates substantially.

In recent years, domain adaptive change detection methods have been proposed to achieve effective change detection in cross-domain scenarios. Zhang et al. [4] presented an adversarial weighted domain adaptation method, which alleviates distribution discrepancies between the source and target domains through adversarial training and a weighting strategy.

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However, when transferring knowledge to the target domain, this method requires retraining the model using relevant target-domain samples, making it unsuitable for rapid deployment to arbitrary unseen target domains.

Recently, several studies [14], [15] have attempted to introduce the DG paradigm into RSCD. Fang et al. [12] proposed an uncertainty-aware DG network that reduces model uncertainty through a dual-covariance style debiasing mechanism and an evidential uncertainty estimation module. Zhao et al. [13] utilized causal disentanglement theory to progressively separate invariant change features from variant domain-specific features, thereby extracting generalized representations.

B. Domain Generalization

To address domain shift, domain adaptation methods have been developed. DT-ST [16], RTea [17], and stable neighbor denoising (SND) [18] enhance model adaptation to the target domain through dynamic teacher updating and category resampling, pseudo-relation learning, and stable-neighborhood denoising, respectively. Nevertheless, these methods generally require access to target-domain samples during training, which limits their applicability when the target domain is unavailable.

In recent years, DG has drawn growing interest. Its primary benefit is the capability to transfer knowledge effectively to entirely new target domains without requiring any target samples during the training phase. This significantly enhances model robustness and applicability in previously unseen scenarios. DG was first introduced in classification tasks, primarily through domain alignment across source domains to learn domain-invariant features [19], [20], meta-learning to acquire generalizable representations [21], or data augmentation to expand source-domain diversity and improve generalization [22].

Subsequently, DG has been extended to semantic segmentation tasks [7], [9], [10], [11]. Since differences in image styles are the primary source of distribution shift, numerous domain-generalized semantic segmentation (DGSS) approaches leverage style information to learn domain-invariant representations. Specifically, feature statistics (e.g., mean, variance, covariance, and Gram matrix) are commonly employed to capture style information [23]. DGSS can be broadly categorized into DR, FN, and vision foundation model (VFM)-based approaches.

DR aims to learn domain-invariant features by diversifying styles in either image space or feature space. For example, Lee et al. [7] utilized ImageNet data [24] as additional data for randomized feature synthesis. Wu et al. [8] employed AdaIN [23] to mix statistics between features and their color-jittered variants, thereby diversifying the trainable feature space. Fan et al. [25] perturb feature statistics to synthesize diverse style domains. Udupa et al. [26] randomly perturb domain-specific fine-grained features while also disturbing coarse-scale style information. However, these methods heavily rely on the diversity of auxiliary domains and often fail to adequately cover unseen styles encountered in real-world scenarios.

Unlike DR, FN instead seeks to remove domain-specific style information from features while retaining domain-invariant content. For example, Pan et al. [9] first proposed a DGSS approach that combines batch normalization [27] and instance normalization [28] within network layers. Batch normalization preserves content information in discriminative features, while instance normalization focuses on removing domain-specific style information. In another study, Choi et al. [10] improved upon previous whitening methods [29], [30] that could lead to content information loss by introducing instance-selective whitening, designed to remove domain-related covariance components. Peng et al. [11] proposed semantic-aware normalization, which performs semantic-aware whitening at the class level and aligns channels via group whitening [29] based on predictions.

More recently, VFMs, which acquire rich and transferable representations through large-scale pretraining, have provided a new direction for DGSS. SeCoV2 [31] leverages interactive segmentation foundation models to construct and refine connectivity-level pseudo-labels for robust semi-supervised DG. FisherTune [32] uses a domain-related Fisher information matrix to identify and selectively update task- and domain-sensitive parameters in VFMs. Nevertheless, retaining the full VFM backbone incurs substantial computational and memory overhead, limiting deployment in resource-constrained scenarios.

C. Contrastive Learning

Contrastive learning seeks to acquire discriminative features by increasing the similarity between positive pairs while reducing the similarity between negative pairs. Recently, contrastive learning has gained significant interest due to its effectiveness in learning discriminative representations across various tasks [33], [34]. van den Oord et al. [35] first introduced the InfoNCE loss, a contrastive loss function designed for self-supervised contrastive learning. Wang et al. [36] introduced class-level contrastive learning to help models learn category-specific embedding spaces. Specifically, they sampled existing categories within an image and conducted contrastive training guided by class labels. Lee et al. [7] applied contrastive learning to learn representations from ImageNet, where ImageNet data was regarded as an auxiliary domain. They applied contrastive learning by treating the stylized wild features and their nearest wild content features as positive pairs.

In RSCD tasks, Wang et al. [37] proposed the scene change contrastive learning framework, which designs global-view and local-view change projections, treating scene changes from the same geographical location as positive pairs and changes from different scene types as negative pairs. Yang et al. [38] introduced cross-domain feature contrastive learning by performing mixed sampling on correctly and incorrectly predicted pixels within deep fused features. Pixels belonging to the same category were regarded as positive examples, whereas those from different categories were considered negative ones, thus improving the separability of ambiguous features. However, these methods focus solely on enhancing interclass discriminability within a single domain,

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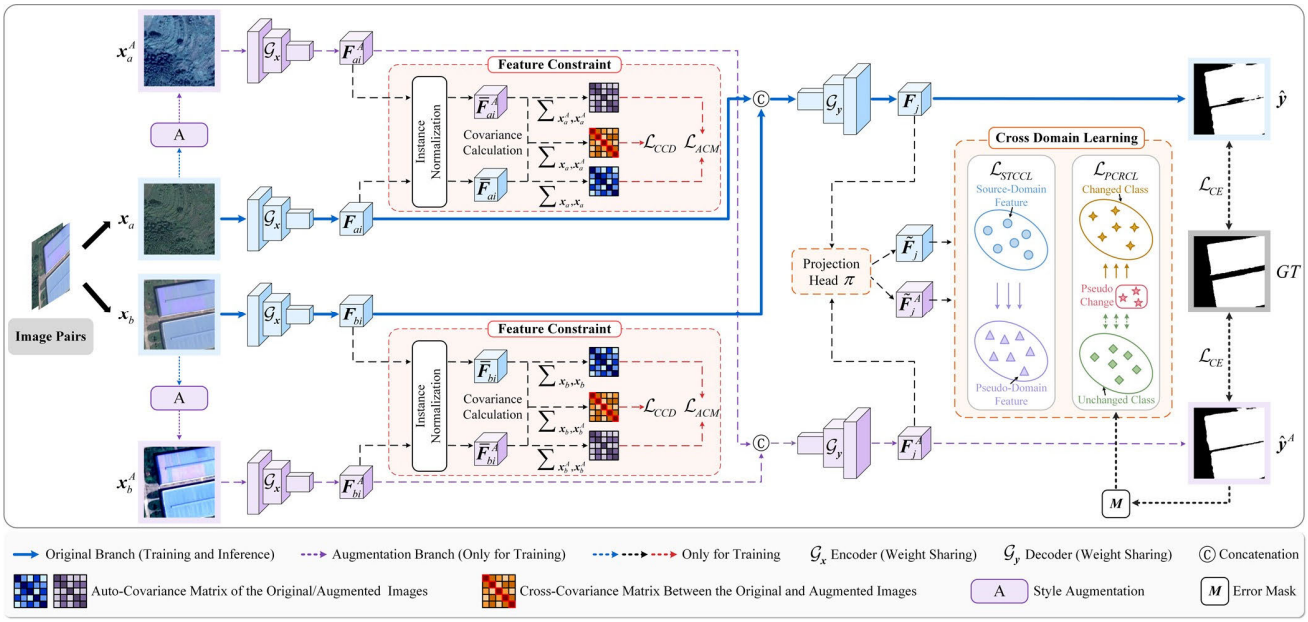


Fig. 2. Architecture of the CDCCNet. The model consists of two branches. The upper branch processes the original source-domain training samples, while the lower branch processes the augmented versions of the source-domain training samples. The FC module performs covariance alignment between the original and augmented images of the same temporal phase to achieve de-stylization. The CDL module samples from the fused feature maps of both original and augmented image pairs for contrastive learning, thereby disentangling similar categories.

while overlooking feature confusion caused by change-domain shifts arising from style discrepancies.

Although existing studies have achieved notable progress in style removal, intradomain contrastive learning, and domain-generalized change detection, several critical issues still remain as follows.

1) Commonly used FN methods, such as normalization and whitening, can effectively eliminate domain-specific styles. However, since image style and semantic content are often highly coupled, these methods inevitably discard content information that is crucial for RSCD.

2) Existing contrastive learning strategies for RSCD typically construct positive and negative sample pairs only within a single domain, making them insufficient for handling cross-domain style shifts caused by different sensors or imaging conditions.

3) Current studies on domain-generalized change detection overlook a critical issue: style discrepancies among images from different datasets can further induce shifts in the change domain, thereby making it difficult for the model to accurately distinguish change categories.

To address the above limitations, we propose an FC module that preserves semantic content information while suppressing the influence of style variations. Furthermore, a CDL module is designed to extend contrastive learning to style-augmented samples. By enforcing semantic consistency among feature representations across different style domains, the proposed module constructs a robust and highly discriminative embedding space, thereby alleviating the impact of change-domain shifts caused by image style discrepancies. As a result, the proposed method fundamentally improves the change detection performance on unseen target domains.

III. Proposed Method

This section first introduces the overall architecture of the proposed method. Then, the two main strategies of CDCCNet are described in detail, followed by a summary of the complete training procedure.

A. Architecture of the Proposed Method

Fig. 2 shows the overall architecture of our proposed method. We independently apply color and texture perturbations to each image in every bi-temporal pair in the training set to generate a new pseudo-domain. Then, four weight-shared encoders are employed to extract bi-temporal features from the original and augmented domains. The bi-temporal features within the same domain are then concatenated to obtain fused features, which are fed into two weight-shared decoders to generate the change maps.

On this basis, we design an FC module, which aligns the feature covariances of identical temporal images from the original and augmented datasets to achieve domain-invariant feature extraction while effectively preserving semantic content. Furthermore, to mitigate the decline in discriminative capability caused by domain shifts, we propose a CDL module, consisting of STCCL and PCRCL, to enhance the model's ability to distinguish change states in fused bi-temporal features.

B. Encoder–Decoder Architecture

As noted previously, both original and augmented image pairs are processed through a shared-weights encoder–decoder network, maintaining a unified pipeline for feature representation and change estimation. First, an encoder built upon

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a convolutional neural architecture extracts feature maps $(\boldsymbol{F}_a, \boldsymbol{F}_b)$ $(\boldsymbol{F}_a^A, \boldsymbol{F}_b^A)$ from the original image pair $(\boldsymbol{x}_a, \boldsymbol{x}_b)$ and its augmented version $(\boldsymbol{x}_a^A, \boldsymbol{x}_b^A)$, respectively. Then, the decoder, implemented with DeepLabV3+ [39], converts the two sets of fused feature maps in parallel into two change prediction maps.

Specifically, the encoder is applied to both the bi-temporal original RS images and the bi-temporal augmented images. The former includes the pre-event original image $\boldsymbol{x}_a \in \mathcal{R}^{H\times W\times 3}$ and the postevent original image $\boldsymbol{x}_b \in \mathcal{R}^{H\times W\times 3}$, from which two independent original feature maps $\boldsymbol{f}_a \in \mathcal{R}^{H'\times W'\times C}$ and $\boldsymbol{f}_b \in \mathcal{R}^{H'\times W'\times C}$ are extracted; the latter includes the pre-event augmented image $\boldsymbol{x}_a^A \in \mathcal{R}^{H\times W\times 3}$ and the postevent augmented image $\boldsymbol{x}_b^A \in \mathcal{R}^{H\times W\times 3}$, from which two independent augmented feature maps $\boldsymbol{f}_a^A \in \mathcal{R}^{H'\times W'\times C}$ and $\boldsymbol{f}_b^A \in \mathcal{R}^{H'\times W'\times C}$ are extracted

$$\begin{cases} \boldsymbol{f}_a = Encoder\left(\boldsymbol{x}_a\right) \\ \boldsymbol{f}_b = Encoder\left(\boldsymbol{x}_b\right) \\ \boldsymbol{f}_d = Concat\left(\boldsymbol{f}_a, \boldsymbol{f}_b\right) \end{cases} \begin{cases} \boldsymbol{f}_a^A = Encoder\left(\boldsymbol{x}_a^A\right) \\ \boldsymbol{f}_b^A = Encoder\left(\boldsymbol{x}_b^A\right) \\ \boldsymbol{f}_d^A = Concat\left(\boldsymbol{f}_a^A, \boldsymbol{f}_b^A\right). \end{cases} \tag{1}$$

This process can be expressed mathematically as (1) where $H \times W$ denotes the spatial dimensions of the image, and C represents the feature dimension. The fused feature map is generated by concatenation along the channel dimension, thereby capturing the key changes between the two temporal states.

Subsequently, the decoder decodes the original fused feature map $\boldsymbol{f}_d$ and the augmented fused feature map $\boldsymbol{f}_d^A$ into the change-activated original logit map $\boldsymbol{p} \in \mathcal{R}^{H\times W\times 2}$ and the augmented logit map $\boldsymbol{p}^A \in \mathcal{R}^{H\times W\times 2}$, respectively, where dimension 2 corresponds to the two classes: change and unchanged. This operation can be expressed as follows:

$$\boldsymbol{p} = Decoder\left(\boldsymbol{f}_d\right), \quad \boldsymbol{p}^A = Decoder\left(\boldsymbol{f}_d^A\right). \tag{2}$$

Finally, the original change logit map $\boldsymbol{p}$ and the augmented logit map $\boldsymbol{p}^A$ are normalized to obtain the original probability map $\hat{\boldsymbol{p}}$ and the augmented probability map $\hat{\boldsymbol{p}}^A$. This process ensures that the probabilities of the change and unchanged classes at each spatial location sum to 1. The process can be expressed as

$$\begin{cases} \hat{\boldsymbol{p}}_{mn} = Soft\max\left(\boldsymbol{p}_{mn}\right) = \dfrac{e^{\boldsymbol{p}_{mn}}}{\sum_{v=1}^{2} e^{\boldsymbol{p}_{mnv}}} \\ \hat{\boldsymbol{p}}_{mn}^A = Soft\max\left(\boldsymbol{p}_{mn}^A\right) = \dfrac{e^{\boldsymbol{p}_{mn}^A}}{\sum_{v=1}^{2} e^{\boldsymbol{p}_{mnv}^A}} \end{cases} \tag{3}$$

where (m, n) denotes the spatial position index.

C. Supervised Learning of the Source Domain

In supervised learning, the RSCD model is trained using all paired samples from the source domain together with their augmented counterparts. For the original RS image pair $(\boldsymbol{x}_a, \boldsymbol{x}_b)$ and the augmented image pair $(\boldsymbol{x}_a^A, \boldsymbol{x}_b^A)$ from the source domain, the two fused feature maps $\boldsymbol{f}_d$ and $\boldsymbol{f}_d^A$ are obtained according to (1), and subsequently transformed into two change probability maps $\hat{\boldsymbol{p}}$ and $\hat{\boldsymbol{p}}^A$ through (2) and (3). The maps $\hat{\boldsymbol{p}}$ and $\hat{\boldsymbol{p}}^A$ are optimized using the cross-entropy loss, which can be mathematically expressed as

$$\begin{cases} \mathcal{L}_{CE}\left(\mathcal{G}_{\boldsymbol{x}}, \mathcal{G}_{\boldsymbol{y}}\right) = \dfrac{1}{H\times W}\displaystyle\sum_{m=1}^{H}\sum_{n=1}^{W} CE\left[\boldsymbol{y}\left(m,n\right), \hat{\boldsymbol{p}}\left(m,n\right)\right] \\ \mathcal{L}_{CE}\left(\mathcal{G}_{\boldsymbol{x}^A}, \mathcal{G}_{\boldsymbol{y}^A}\right) = \dfrac{1}{H\times W}\displaystyle\sum_{m=1}^{H}\sum_{n=1}^{W} CE\left[\boldsymbol{y}\left(m,n\right), \hat{\boldsymbol{p}}^A\left(m,n\right)\right]. \end{cases} \tag{4}$$

The final total cross-entropy loss is defined as

$$\mathcal{L}_{CE} = \left(\mathcal{L}_{CE}\left(\mathcal{G}_{\boldsymbol{x}}, \mathcal{G}_{\boldsymbol{y}}\right) + \mathcal{L}_{CE}\left(\mathcal{G}_{\boldsymbol{x}^A}, \mathcal{G}_{\boldsymbol{y}^A}\right)\right)/2. \tag{5}$$

This enables the RSCD model to learn from the labeled data in the source domain and acquire preliminary change detection capability. However, due to domain shift, the models cannot directly adapt to unseen target domains in the absence of DG.

D. FC Module

In RSCD, variations in the visual characteristics (i.e., styles) of images from different data sources lead to domain shifts. IBN-Net [9] indicates that style-induced shifts in feature representations mainly emerge within the initial layers of deep networks. The IBN-Net [9] further reveals that the shallow feature distributions can reflect the style information of input images, and these shallow layers correspond to the encoder of segmentation models. RobustNet [10] has also shown that feature covariance contains domain-specific style information, such as texture and color.

It is a fact that style manifests not only in the distinct color palettes of different images but also in their textures and spatial structures. The covariance matrix of image features effectively captures image texture by quantifying correlations between channels. Previous work typically used whitening transformations to remove differences in image texture, which, while removing style information, often also removes some key content.

Based on these insights, our method addresses style variations in the encoder by introducing the proposed ACML and CCDL. As noted in IBN-Net [9], early layers tend to encode style information; therefore, we apply the ACML and CCDL losses only to the input convolutional layer as well as to the ResNet Stage 1 and 2.

1) Auto-Covariance Matching Loss: To enable the network to consistently adapt to diverse visual styles while preserving semantic content, the ACML is proposed. Specifically, this loss minimizes the discrepancy between the auto-covariance matrices of features extracted from different styled images corresponding to the same temporal phase. Given a pair of images $(\boldsymbol{x}_k, \boldsymbol{x}_k^A)$ from the kth temporal phase, the features extracted by the ith block of the encoder are denoted as $\boldsymbol{F}_{ki} \in \mathcal{R}^{(H^{ki}\times W^{ki})\times C^{ki}}$ and $\boldsymbol{F}_{ki}^A \in \mathcal{R}^{(H^{ki}\times W^{ki})\times C^{ki}}$, respectively, where H^{ki}, W^{ki}, and C^{ki} represent the height, width, and channel number of the feature map, respectively. Moreover, following the approach in [5], [10], we compute covariance matrices using features processed by instance normalization [28] to ensure scale consistency

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across features. The feature maps are normalized and flattened into $\bar{\boldsymbol{F}}_{ki}$ and $\bar{\boldsymbol{F}}_{ki}^{A} \in \mathcal{R}^{(H^{ki}W^{ki})\times C^{ki}}$, which are formally defined as follows:

$$\begin{cases} \bar{\boldsymbol{F}}_{ki} = \dfrac{(\boldsymbol{F}_{ki} - \mu(\boldsymbol{F}_{ki}))}{\sigma(\boldsymbol{F}_{ki}) + \varepsilon} \\ \bar{\boldsymbol{F}}_{ki}^{A} = \dfrac{\left(\boldsymbol{F}_{ki}^{A} - \mu\left(\boldsymbol{F}_{ki}^{A}\right)\right)}{\sigma\left(\boldsymbol{F}_{ki}^{A}\right) + \varepsilon} \end{cases} \tag{6}$$

where $\mu(\cdot) \in \mathcal{R}^{C^{ki}}$ and $\sigma(\cdot) \in \mathcal{R}^{C^{ki}}$ denote the channel-wise mean and standard deviation, respectively, computed over the spatial dimensions of each feature map, and $\varepsilon > 0$ is a small positive constant introduced to prevent division by zero and improve numerical stability. The normalized features corresponding to the same temporal phase are then used to compute the auto-covariance matrices of the original and augmented image features, denoted by $\sum^{i}_{\boldsymbol{x}_k,\boldsymbol{x}_k} \in \mathcal{R}^{C^{ki}\times C^{ki}}$ and $\sum^{i}_{\boldsymbol{x}_k^A,\boldsymbol{x}_k^A} \in \mathcal{R}^{C^{ki}\times C^{ki}}$, respectively, as in the following equation:

$$\begin{cases} \displaystyle\sum_{\boldsymbol{x}_k,\boldsymbol{x}_k}^{i} = \frac{1}{H^{ki}W^{ki}-1}\left(\bar{\boldsymbol{F}}_{ki}\right)^{T} \cdot \bar{\boldsymbol{F}}_{ki} + \varepsilon \boldsymbol{I}_{C_i} \\ \displaystyle\sum_{\boldsymbol{x}_k^A,\boldsymbol{x}_k^A}^{i} = \frac{1}{H^{ki}W^{ki}-1}\left(\bar{\boldsymbol{F}}_{ki}^{A}\right)^{T} \cdot \bar{\boldsymbol{F}}_{ki}^{A} + \varepsilon \boldsymbol{I}_{C_i} \end{cases} \tag{7}$$

where $\boldsymbol{I}_{C_i} \in \mathcal{R}^{C^{ki}\times C^{ki}}$ denotes the identity matrix. Subsequently, the ACML is designed to align these auto-covariance matrices, thereby ensuring that the network maintains feature consistency under different style variations. The ACML is defined as follows:

$$\mathcal{L}_{ACM} = \frac{1}{n_e\left(C^{ki}\right)^2}\sum_{k=1}^{2}\sum_{i=1}^{n_e}\left\|\sum_{\boldsymbol{x}_k,\boldsymbol{x}_k}^{i} - \sum_{\boldsymbol{x}_k^A,\boldsymbol{x}_k^A}^{i}\right\|_2 \tag{8}$$

where n_e denotes the number of blocks in the encoder.

2) Cross-Covariance Diagonal Loss: Although the ACML enforces consistency between the feature distributions of images with different styles but identical content, it does not fully account for the correlations among feature channels between the two images. To address this limitation, we introduce the CCDL, which aims to achieve content alignment by enforcing consistent channel responses between the image pair $(\boldsymbol{x}_k, \boldsymbol{x}_k^A)$ with different styles but identical content through their cross-covariance matrix. Given a pair of normalized feature maps $(\bar{\boldsymbol{F}}_{ki}, \bar{\boldsymbol{F}}_{ki}^{A})$, the cross-covariance matrix between the original and augmented features, denoted by $\sum^{i}_{\boldsymbol{x}_k,\boldsymbol{x}_k^A} \in \mathcal{R}^{C^{ki}\times C^{ki}}$, is computed as

$$\sum_{\boldsymbol{x}_k,\boldsymbol{x}_k^A}^{i} = \frac{1}{H^{ki}W^{ki}-1}\left(\bar{\boldsymbol{F}}_{ki}\right)^{T} \cdot \bar{\boldsymbol{F}}_{ki}^{A} + \varepsilon \boldsymbol{I}_{C_i}. \tag{9}$$

The cross-covariance matrix is expected to approximate the identity matrix, as the feature pair should encode the same content information. However, some critical content information is represented through cross-channel interactions and cannot be simply ignored by setting the off-diagonal elements of the cross-covariance matrix to zero. Therefore, the proposed CCDL only enforces the diagonal elements of the cross-covariance matrix to converge to 1. This approach mitigates the drawback of existing FN methods [10], [11], which tend to remove important content information. The CCDL is defined as

$$\mathcal{L}_{CCD} = \frac{1}{n_e C^{ki}}\sum_{k=1}^{2}\sum_{i=1}^{n_e}\left\|diag\left(\sum_{\boldsymbol{x}_k,\boldsymbol{x}_k^A}^{j}\right) - \mathbf{1}\right\|_2 \tag{10}$$

where $diag\left(\sum_C\right) \in \mathcal{R}^C$ denotes the column vector containing the diagonal elements of $\sum^{i}_{\boldsymbol{x}_k,\boldsymbol{x}_k^A}$, and $\mathbf{1} \in \mathcal{R}^C$ denotes a column vector with all elements equal to 1.

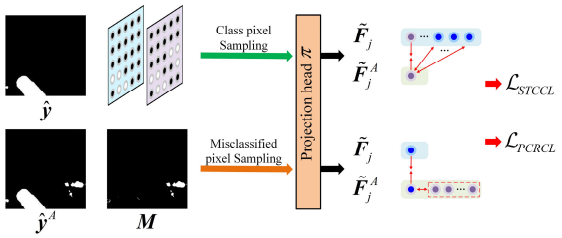


Fig. 3. Illustration of the CDL module: the mask M represents the error mask obtained from the change prediction map of the augmented image pair. STCCL performs contrastive learning based on sampling from different spatiotemporal change categories, whereas PCRCL conducts contrastive learning under the guidance of M. Both methods share a projection head π to generate semantic representations.

E. CDL Module

In fact, for RSCD, domain shifts not only include feature discrepancies between images acquired at different temporal phases, but also encompass domain shifts across different datasets, namely, change-domain shifts. Due to the influence of image style discrepancies among different datasets, the features corresponding to different change states may further become highly similar, thereby aggravating change-domain shifts. For the decoder, we adopt a contrastive learning approach, which has been shown to be effective in extracting discriminative features [33]. Specifically, we employ the InfoNCE loss function [35], which is defined as follows:

$$\mathcal{L}_{IN}(a,p,n) = -\log\left(\frac{e^{(a\cdot p/\tau)}}{e^{(a\cdot p/\tau)} + \sum_{n}^{N-} e^{(a\cdot n/\tau)}}\right) \tag{11}$$

where a denotes the anchor, p denotes the positive sample, n denotes the negative sample, and $N-$ represents the set of negative samples.

To achieve consistency of feature representations across different styles, we introduce a CDL module. Specifically, the anchor is derived from the fused features of the augmented image pair, while the positive sample is extracted from the same pixel location in the fused features of the corresponding original image pair. As illustrated in Fig. 3, our method primarily consists of two core losses: STCCL and PCRCL.

1) Spatio-Temporal Consistency Contrastive Learning: STCCL aims to construct a discriminative embedding space for each category by treating pixels from different categories in the fused features of the original image pairs as negative samples, thereby learning more robust feature representations. For the original bi-temporal image pair $(\boldsymbol{x}_a, \boldsymbol{x}_b)$ and the augmented bi-temporal image pair $(\boldsymbol{x}_a^A, \boldsymbol{x}_b^A)$, we consider the fused feature outputs from the jth layer of the decoder. For a pixel

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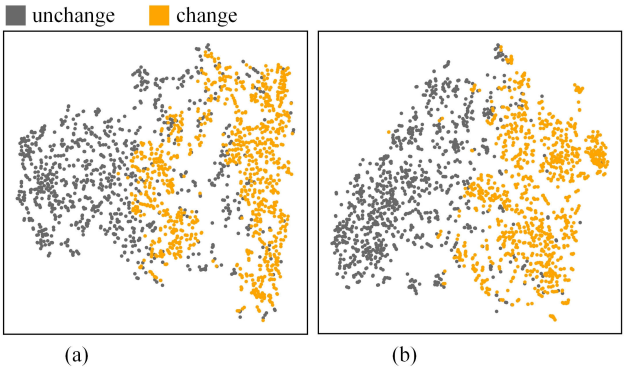


Fig. 4. t-SNE [48] visualization compares the cases with and without PCRCL. As shown in (b), after applying PCRCL, the features of the change and nonchange categories exhibit a clear separation. (a) Without $\mathcal{L}_{PCRCL}$. (b) With $\mathcal{L}_{PCRCL}$.

at position (m, n), the fused features are denoted as $\boldsymbol{F}_j(m, n)$ and $\boldsymbol{F}_j^A(m, n) \in \mathcal{R}^{(1\times1)\times C^j}$, where C^j represents the channel dimension of the fused features at that layer. We treat $\boldsymbol{F}_j^A(m, n)$ as the anchor and $\boldsymbol{F}_j(m, n)$ as the positive sample, because they correspond to the same spatial location and contain identical semantic content. To obtain negative samples, we downsample the change label map $\boldsymbol{y}_j \in \mathcal{R}^{(H^j\times W^j)\times 2}$ and select pixels belonging to categories different from that of the anchor pixel as negative samples. Before being passed into the contrastive loss, all fused features are processed through a projection head, denoted as π, to produce the projected fused features $\tilde{\boldsymbol{F}}_j$ and $\tilde{\boldsymbol{F}}_j^A$. Our STCCL loss is defined as follows:

$$\mathcal{L}_{STCCL} = \sum_{j}^{n_d} \mathcal{L}_{IN}\left(\tilde{\boldsymbol{F}}^A_{j,(m,n)}, \tilde{\boldsymbol{F}}_{j,(m,n)}, \tilde{\boldsymbol{F}}_{j,(p,q)}\right), \quad \text{where}$$
$$(p, q) \in \left\{(p, q) \in P | \boldsymbol{y}_{j,(p,q)} \neq \boldsymbol{y}_{j,(m,n)}\right\}. \tag{12}$$

The set P denotes all pixel locations in the fused feature map $\boldsymbol{F}_j$, with dimensions of $H^j \times W^j$, corresponding to the height and width of the fused feature map, respectively. In addition, n_d represents the total number of blocks in the decoder.

2) Pseudo-Change Rectification Contrastive Learning: In RSCD, style discrepancies among images from different datasets can further induce shifts in the change domain, which may lead to feature confusion between similar changed and unchanged categories, thereby resulting in misclassification, as illustrated in Fig. 4. To alleviate this problem, we introduce PCRCL, which is specifically designed to separate misclassified features in the fused representation of augmented image pairs, bringing them closer to the correct category while pushing them away from the misclassified category, thereby achieving effective feature disentanglement. To further ensure consistency in the feature space and to capture discriminative information, the projection head π used in STCCL is shared. The prediction map for each augmented image pair, denoted by $\hat{\boldsymbol{y}}^A = \phi(\boldsymbol{x}_a^A, \boldsymbol{x}_b^A)$, is resized to $\hat{\boldsymbol{y}}^A_{j,(m,n)} \in \mathcal{R}^{(H^j\times W^j)\times 2}$. Similarly, $\boldsymbol{y}_{j,(m,n)}$ represents the corresponding ground-truth label maps. Anchor points are established at a position $\hat{\boldsymbol{y}}^A_{j,(m,n)} \neq \boldsymbol{y}_{j,(m,n)}$ using these predictions and label maps. Negative examples are drawn from the fused features of augmented images associated with the anchor point's misclassified category. All samples are then passed through the projection head π. Our PCRCL loss function is defined as follows:

$$\mathcal{L}_{PCRCL} = \sum_{j}^{n_d} \mathcal{L}_{IN}\left(\tilde{\boldsymbol{F}}^A_{j,(m,n)}, \tilde{\boldsymbol{F}}_{j,(m,n)}, \tilde{\boldsymbol{F}}^A_{j,(r,s)}\right), \quad \text{where}$$
$$(r, s) \in \left\{(r, s) \in P | \boldsymbol{y}_{j,(r,s)} = \hat{\boldsymbol{y}}_{j,(m,n)}\right\}. \tag{13}$$

Finally, by combining the cross-entropy loss $\mathcal{L}_{CE}$ with the other loss terms, the overall loss function is defined as

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \omega_1\mathcal{L}_{ACM} + \omega_2\mathcal{L}_{CCD} + \omega_3\mathcal{L}_{STCCL} + \omega_4\mathcal{L}_{PCRCL} \tag{14}$$

where ω_1, ω_2, ω_3, and ω_4 denote the weighting factor assigned to each loss function.

IV. Experiments

In RSCD, synthetic data are easier to obtain and annotate compared to real-world data, making them more suitable for practical applications. Therefore, we train our model on a simulated dataset and evaluate its cross-DG capability on real-world datasets. This section details the experimental implementation, the design of comparative experiments with existing DG methods, and the ablation studies conducted to verify the effectiveness of our approach. In addition, we assess the robustness and applicability of the method in cross-domain scenarios where training is performed on one real dataset and testing on other real datasets.

A. Datasets

In the experiment, we trained the model using a synthetic dataset (SyntheWorld [40]) designed for building change detection, then tested it on three real-world image datasets (GZ [41], WHU [42], and EGY [43]) for change detection to evaluate cross-domain applicability. Specifically, we assessed improvements in model generalization performance when applying synthetic data to real data to evaluate the effectiveness of our proposed method.

1) Synthetic Dataset: SyntheWorld is a large-scale, automatically annotated dataset built through the integration of procedural 3-D modeling and artificial intelligence-generated content (AIGC). It is specifically designed for land cover mapping and building RSCD tasks. The dataset contains 40 000 pairs of bi-temporal images with annotated building changes. Among them, 30 000 pairs simulate relatively structured urban and suburban layouts, with image dimensions of 512 × 512 pixels and a spatial resolution of 0.3–0.6 m/pixel. These image pairs are cropped into nonoverlapping 256 × 256 patches, from which 10 000 pairs are randomly selected for training and 2000 pairs for testing. The resulting subset has a change-pixel ratio of 8.16% and covers diverse buildings of various types, colors, sizes, and functions, demonstrating high diversity and representativeness.

2) Real-World Datasets: GZ is composed of VHR satellite imagery covering suburban areas of Guangzhou, China, spanning from 2006 to 2019, and is used for monitoring urban expansion. Acquired via Google Earth services, this

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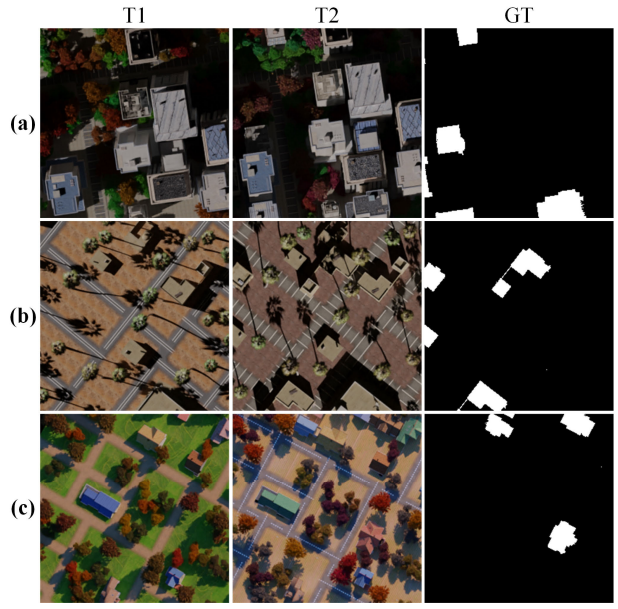


Fig. 5. Sample examples from the SyntheWorld dataset. (a) Urban style. (b) Desert style. (c) Suburban or rural style.

dataset contains 19 pairs of seasonal images with a spatial resolution of 0.55 m/pixel and a change-pixel ratio of 8.5%. They are divided into training/validation/test subsets of 2161/720/720 pairs following a 6:2:2 ratio. WHU is derived from aerial imagery, including land cover changes caused by the reconstruction following the Christchurch earthquake in New Zealand, with a temporal span from 2012 to 2016. It contains a pair of RS images with dimensions of $32\,507 \times 15\,354$ pixels and a spatial resolution of 0.075 m/pixel, with a change-pixel ratio of 4.33%. The original images are divided into nonoverlapping 256×256 patches, which are further split into 5947 training images, 743 validation images, and 744 test images. EGY consists of high-resolution satellite imagery covering four Egyptian cities (New Mansoura, El Galala City, New Cairo, and New Thebes) over the period 2015–2022. It contains 4263 training pairs, 1218 validation pairs, and 610 test pairs, with a spatial resolution of 0.25 m/pixel and a change-pixel ratio of 6.97%.

From the perspective of domain shift, significant differences are evident not only between the synthetic dataset and real-world datasets but also within the synthetic dataset itself and among the real-world datasets. These differences include variations in spatial resolution (e.g., SyntheWorld at 0.3–0.6 m/pixel, GZ at 0.55 m/pixel, and WHU at 0.075 m/pixel) as well as regional differences (SyntheWorld not being tied to a specific region, WHU representing Christchurch, New Zealand, GZ representing Guangzhou, China, and EGY covering four Egyptian cities). Such differences highlight the diversity of change semantics across datasets.

It is worth noting that the SyntheWorld dataset is deliberately designed to simulate representative styles of different real-world regions, such as high-rise urban areas, low-rise suburban neighborhoods, grasslands and farmlands, as well as mountainous and desert regions characterized by earthen houses and bare land, as illustrated in Fig. 5. These stylistically diverse images form distinct clusters in the feature space, exhibiting considerable domain shift.

B. Evaluation Metrics

In our experiments, we utilize three standard metrics to evaluate the performance of all methods: $F1$-score ($F1$), intersection over union (IoU), and the Kappa coefficient (Kappa). The values of IoU and $F1$ span 0%–100%, whereas the Kappa coefficient lies between -1 and 1. For each metric, higher values correspond to superior change detection performance. The corresponding formulas are given below

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{15}$$

$$IoU = \frac{TP}{(TP + FP + FN)} \tag{16}$$

$$OA = \frac{TP + TN}{FP + FN + TP + TN} \tag{17}$$

$$PRE = \frac{(TN + FN) \times (FP + TN)}{(TN + TP + FN + FP)^2} + \frac{(TP + FP) \times (FN + TP)}{(TN + TP + FN + FP)^2} \tag{18}$$

$$Kappa = \frac{OA - PRE}{1 - PRE} \tag{19}$$

where TN denotes the number of pixels correctly identified as no-change, and TP denotes the number of pixels correctly identified as change. Conversely, FN represents the number of change pixels misclassified as no-change, and FP represents the number of no-change pixels misclassified as change. As this work concentrates on change regions, only metrics pertaining to the change class are computed and presented.

C. Implementation Details

All experiments were conducted using PyTorch 2.2.1 and executed on a single NVIDIA RTX 4090 GPU. During training, all methods adopt DeepLabV3+ [39] as the RSCD network, and use ResNet-50 [44], MobileNetV2 [45], and ShuffleNetV2 [46] pretrained on ImageNet [24] as the shared-weight Siamese encoders to extract bi-temporal features, with a spatial scaling factor s set to 16. The bi-temporal features are then fused and fed into the decoder. In addition, all methods employ the per-pixel auxiliary cross-entropy loss proposed in PSPNet [47], and the fused shallow bi-temporal features from the ResNet Stage 1 are concatenated with the deep fused features of the decoder.

To ensure a fair comparison, all experiments are repeated three times, and the average results are reported. All models are trained under identical conditions using the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of 0.0005, for 50 epochs with a batch size of 8. A polynomial learning rate scheduler is used with an initial learning rate of 0.01 and a power of 0.9.

During training, color and texture perturbations are independently applied to each image in a bi-temporal pair $(\boldsymbol{x}_a, \boldsymbol{x}_b)$ with a probability of 1.0. Although both images use the same perturbation types and parameter ranges, their parameters are sampled independently to simulate appearance variations

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TABLE I

QUANTITATIVE COMPARISON RESULTS USING RESNET-50/MOBILENETV2/SHUFFLENETV2. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Trained on SyntheWorld													
Tested on		GZ			WHU			EGY			SyntheWorld		
Backbone	Methods	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
ResNet-50	Baseline [39]	45.61	29.67	40.45	38.86	24.45	34.20	40.00	25.04	35.52	88.97	80.13	88.01
	IBN-Net [9]	43.97	28.31	39.70	45.94	29.79	42.94	35.80	21.88	31.70	89.65	81.25	88.76
	RobustNet [10]	49.45	33.02	44.46	45.73	29.71	42.29	38.88	24.18	33.78	88.44	79.27	87.43
	WildNet [7]	49.22	32.77	44.35	46.11	30.04	42.60	33.21	19.97	28.04	87.02	77.03	85.88
	SANSAW [11]	43.98	28.24	38.97	29.93	17.79	24.21	39.67	24.80	33.98	89.42	80.87	88.48
	NP+ [25]	48.14	31.82	43.69	46.59	30.38	43.47	45.94	29.83	41.70	89.92	81.69	89.03
	MRFP+ [26]	46.89	30.68	42.31	44.46	28.60	41.28	39.58	24.68	34.97	88.46	79.32	87.48
	Ours	**56.33**	**39.25**	**52.37**	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**	**89.94**	**81.72**	**89.06**
MobileNetV2	Baseline	44.94	29.20	40.34	44.01	28.22	40.35	38.45	23.81	32.09	87.31	77.48	86.20
	IBN-Net	42.63	27.32	37.95	43.26	25.67	37.29	37.71	23.34	33.26	88.15	78.81	87.11
	RobustNet	45.02	29.10	41.26	43.46	27.77	40.59	41.29	26.02	37.98	87.79	78.25	86.73
	NP+	22.52	12.69	19.08	47.57	31.25	45.15	34.47	20.84	30.14	87.08	77.12	85.97
	MRFP+	40.06	25.05	36.30	**48.17**	**31.73**	**45.70**	32.42	19.37	28.85	85.65	74.89	84.46
	Ours	**52.08**	**35.31**	**48.13**	46.95	30.69	44.06	**44.10**	**28.35**	**40.30**	**88.49**	**79.38**	**87.49**
ShuffleNetV2	Baseline	36.07	22.12	31.16	40.89	25.79	37.45	31.27	18.75	27.93	85.19	74.20	83.91
	IBN-Net	34.08	25.03	29.23	35.34	21.51	31.95	37.17	22.82	32.72	85.53	74.73	84.27
	RobustNet	45.85	29.80	39.39	37.24	22.94	32.80	27.16	15.77	20.38	84.80	73.67	83.50
	NP+	39.87	24.92	34.93	36.16	22.13	32.93	36.69	22.49	32.21	84.87	73.72	83.55
	MRFP+	28.90	17.06	24.54	26.55	15.42	23.52	28.14	16.48	24.37	84.08	72.53	82.74
	Ours	**51.31**	**34.54**	**46.24**	**43.97**	**28.19**	**40.53**	**43.59**	**27.88**	**38.81**	**85.92**	**75.31**	**84.70**

caused by differences in acquisition time, sensor conditions, and environmental factors. Color perturbations include brightness, contrast, saturation, and hue adjustments, with the first three factors uniformly sampled from ([0.7, 1.3]) and the hue shift from ([−0.1, 0.1]). Texture perturbation is implemented using Gaussian blur, whose standard deviation is uniformly sampled from ([0.15, 1.30]). The weighting parameters ω_1, ω_2, ω_3, and ω_4 in (14) are set to 0.2, 0.2, 0.3, and 0.3, respectively.

D. Comparison Experiments

To objectively evaluate the effectiveness of the proposed CDCCNet, we compared it with seven typical methods, including the baseline and six state-of-the-art DG techniques, across four scenarios: SyntheWorld → EGY, SyntheWorld → GZ, SyntheWorld → WHU, and SyntheWorld → SyntheWorld. All methods used the same encoder–decoder architecture. In particular, WildNet utilizes external datasets (e.g., ImageNet) to expand the style and content information of the source domain; therefore, in our reproduction, we replaced its external datasets with the source dataset to ensure a fair comparison.

Baseline [39] uses only the entire training dataset of the source domain for supervised training and serves as the baseline for all subsequent methods. It employs the DeepLabV3+ network with ResNet50, MobileNetV2, or ShuffleNetV2 as the backbone.

The six methods for comparison are IBNNet [9], RobustNet [10], WildNet [7], SANSAW [11], NP+ [25], and MRFP+ [26], respectively. All these methods possess distinctive characteristics, such as proposing a significant model for the first time or demonstrating the best performance. They include current typical DG approaches and provide a comprehensive reference for the method proposed in this article.

Furthermore, due to experimental constraints, quantitative comparisons with two recent domain-generalized change detection methods, namely, GenerChange [12] and CDDGNet [13], were conducted only under their respective experimental settings.

1) Quantitative Analysis: Table I presents the cross-domain quantitative evaluation results of our proposed method and other comparison methods trained on SyntheWorld using different backbone networks. First, when trained with ResNet-50, our proposed method outperforms all other approaches. Compared with FN-based methods that remove domain-specific styles, our approach minimizes the loss of essential content information to the greatest extent. Specifically, in the SyntheWorld → GZ scenario, our method achieves improvements of 6.88%, 6.23%, and 7.91% in $F1$-score, IoU, and Kappa coefficient, respectively, over the second-best method, and improvements of 10.72%, 9.58%, and 11.92% over the baseline. In the SyntheWorld → WHU scenario, it achieves improvements of 3.54%, 3.10%, and 3.60% over the second-best method, and 11.27%, 9.03%, and 12.87% over the baseline. For the SyntheWorld → EGY scenario, improvements are 2.79%, 2.45%, and 3.16% compared with the second-best method, and 8.73%, 7.24%, and 9.34% over the baseline. Notably, our method also improves the source domain performance, achieving relative gains of 0.97%, 1.59%, and 1.05% over the baseline. Overall, these quantitative results across three cross-domain scenarios demonstrate that our method effectively enhances DG capability.

To test whether the performance advantages of our method are universal across different backbone networks, we further tested it using the MobileNetV2 and ShuffleNetV2 as the backbone. The experimental results are shown in Table I. It can be seen that our method also performs excellently in

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both source and unseen target domains when trained with the MobileNetV2, achieving the best or second-best results across all metrics. Specifically, in the SyntheWorld → GZ scenario, $F1$, IoU, and Kappa coefficients are improved by 7.06%, 6.21%, and 6.87% compared with the second-best method, and by 7.14%, 6.11%, and 7.79% compared with the baseline. In the SyntheWorld → WHU scenario, the results are only 1.22%, 1.04%, and 1.64% lower than the best method, but still 2.94%, 2.47%, and 3.71% higher than the baseline. For the SyntheWorld → EGY scenario, improvements over the second-best method reach 2.81%, 2.33%, and 2.32%, while gains over the baseline are 5.65%, 4.54%, and 8.21%. Furthermore, our method improves the source domain results by 1.18%, 1.9%, and 1.29% compared to the baseline.

When trained with ShuffleNetV2, our method consistently outperforms all comparison methods in both source and unseen target domains. Specifically, in the SyntheWorld → GZ scenario, our method achieves 5.46%, 4.74%, and 6.85% higher $F1$, IoU, and Kappa coefficients than the second-best method, and 15.24%, 12.42%, and 15.08% improvements over the baseline. In the SyntheWorld → WHU scenario, improvements over the baseline reach 3.08%, 2.40%, and 3.08%, while in the SyntheWorld → EGY scenario, the gains over the second-best method are 6.42%, 5.06%, and 6.09%, and 12.32%, 9.13%, and 10.88% over the baseline. Furthermore, our method improves the source domain results by 0.73%, 1.11%, and 0.79% compared to the baseline.

Although some competing methods, such as NP+, RobustNet, and MRFP+, occasionally demonstrate good performance, their results are unstable. For example, when using ResNet-50, NP+ achieved IoUs of 31.82% and 29.83% on the SyntheWorld → GZ and SyntheWorld → EGY cross-domain scenarios, respectively, improving over the baseline by 2.15% and 4.79%. However, when trained with MobileNetV2, its DG performance dropped significantly, with IoUs of only 12.69% and 20.84%, which are 16.51% and 2.97% lower than the baseline. In contrast, our proposed method consistently achieves superior robustness and generalization capability under all experimental conditions. Except for the SyntheWorld → WHU scenario using MobileNetV2, where it is slightly lower than the best-performing method, our approach surpasses all others across all settings.

Table II presents a quantitative comparison between CDCCNet and the latest domain generalization methods for change detection. Under the experimental settings and evaluation protocols adopted in GenerChange, the proposed method achieves the best performance in the vast majority of cross-domain scenarios. Specifically, CDCCNet improves mIoU by up to 5.16% compared with the second-best method and achieves an improvement of up to 4.85% in mF1. Notably, it ranks slightly below the best-performing method in only one cross-domain scenario. These results indicate that the proposed method can effectively alleviate the performance degradation caused by image style discrepancies and the resulting change-domain shifts in RSCD, while exhibiting stronger DG capability and robustness.

2) Qualitative Analysis: To qualitatively demonstrate the superior generalization capability of our proposed method, we further compared its RSCD results with those of other approaches. All methods used ResNet-50 as the backbone, were trained on the synthetic dataset SyntheWorld, and tested on three real-world datasets (GZ, WHU, and EGY).

Fig. 6 presents a visual comparison of our method and other approaches in six representative change scenarios under the SyntheWorld → GZ cross-domain setting. Compared to the synthetic (source) dataset, the bi-temporal images in the GZ dataset exhibit more complex imaging conditions and a greater variety of pseudo-changes. For large-scale building changes [see Fig. 6(a), (c), and (d)], other methods show significant omissions and generate false detections in pseudo-change regions. In contrast, our method only produces minor red and green regions in small areas. Fig. 6(b) and (f) demonstrates that our method handles subtle changes and occlusions more effectively, avoiding false positives and false negatives even under complex background interference. Due to high similarity between the change targets and the background, omissions occur in the results of other methods in Fig. 6(e); our method, however, successfully distinguishes similar classes, effectively reducing missed detections. From a boundary perspective, our method preserves more complete and continuous building contours, with substantially less boundary erosion, expansion, and fragmentation than the comparison methods.

We visualized the results of all methods in the SyntheWorld → WHU cross-domain scenario, as shown in Fig. 7. Compared to the synthetic (source) dataset, the WHU bi-temporal images exhibit larger optical differences, with clearer object textures. The greater color variations cause the baseline and other methods to produce omissions and false alarms when predicting changes. Our method, however, effectively adapts to such style differences, generating more accurate change maps, as illustrated in Fig. 7(a) and (b). Fig. 7(c)–(f) shows that other methods incorrectly detect changes from road/land to cement surfaces in the background. Our method, by learning more discriminative semantic features, can distinguish highly similar surface information and effectively suppress such false positives. At the boundary level, our method better maintains the regular geometric structures and edge continuity of the changed buildings, while reducing the boundary deviations caused by over-detection and omission in the comparison methods.

To more intuitively illustrate our method's advantages on the EGY dataset, we present representative test cases in Fig. 8. Compared to the synthetic (source) dataset, the EGY scenes focus on building changes in desert cities. In Fig. 8(a), spatial offsets, shadow changes, and similar surface information in multitemporal images cause significant interference for other methods, while our method maintains strong generalization and robustness. Fig. 8(b), (c), and (e) demonstrates that our method achieves superior detection performance for dense building changes at multiple scales, preserving the edges of dense change regions more completely. For buildings with desert-like colors and textures similar to the prechange region [see Fig. 8(f)], our method effectively identifies the changes. Overall, this performance is mainly attributed to our method's ability to extract domain-invariant features and effectively distinguish cross-class semantic information, enhancing the model's generalization to both style and change diversity. Moreover, at the boundary level, our method better preserves

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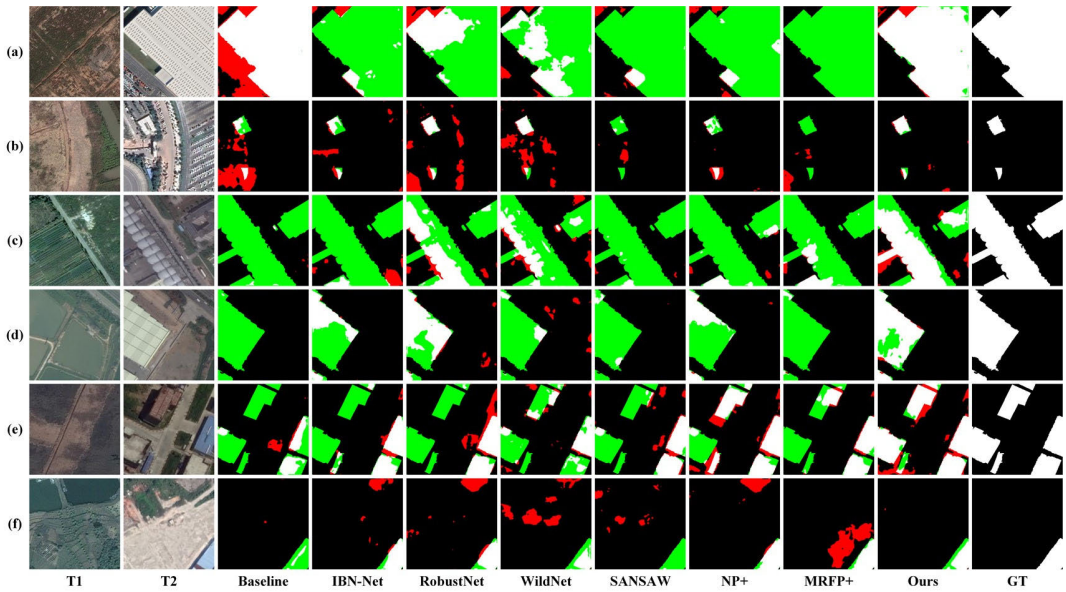


Fig. 6. Qualitative comparison between methods trained on SyntheWorld and tested on unseen target domains of GZ using DeepLabV3+ with a ResNet-50 backbone. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (a)–(f) Six representative change detection examples, including the bitemporal images, predictions of different methods, and the corresponding ground truth.

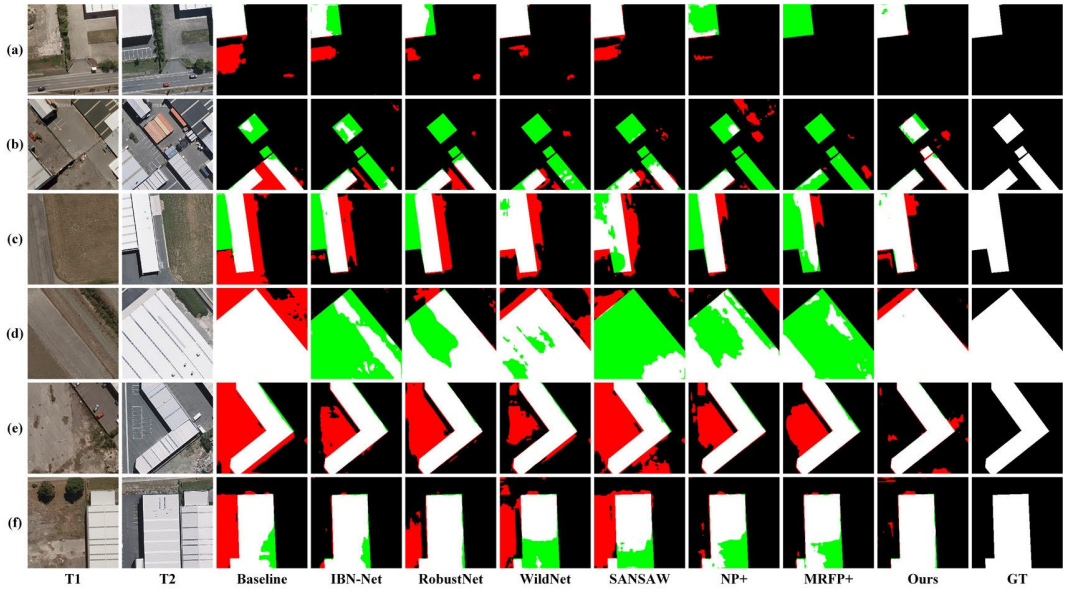


Fig. 7. Qualitative comparison between methods trained on SyntheWorld and tested on unseen target domains of WHU using DeeplabV3+ with ResNet-50 backbone. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (a)–(f) Six representative change detection examples, including the bitemporal images, predictions of different methods, and the corresponding ground truth.

TABLE II

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART DOMAIN-GENERALIZED CHANGE DETECTION METHODS. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

	LEVIR → WHU	LEVIR → GZ	LEVIR+ → WHU	WHU → LEVIR	WHU → GZ	WHU → EGY
Methods	mIoU/mF1/OA	mIoU/mF1/OA	mIoU/mF1/OA	mIoU/mF1/OA	mIoU/mF1/OA	mIoU/mF1/OA
GenerChange [12]	77.22/85.64/97.63	63.87/74.05/92.98	-	54.02/60.15/95.22	**66.16**/**76.67**/92.32	54.81/62.60/92.77
CDDGNet [13]	65.94/79.47/**98.21**	-	70.93/82.99/**98.52**	-	-	-
Ours	**78.04**/**86.29**/97.70	**65.36**/**75.36**/**94.03**	**76.09**/**84.71**/97.65	**57.35**/**65.00**/**95.44**	65.60/75.61/**94.06**	**55.18**/**62.68**/**93.73**

the contours of densely distributed buildings and the narrow separations between adjacent objects, thereby reducing boundary fragmentation and the merging of neighboring change regions.

E. Ablation Study

1) Selection of FC Layers: Earlier-stage feature maps have larger spatial dimensions and contain more low-level details, whereas later-stage feature maps, although lower in resolution,

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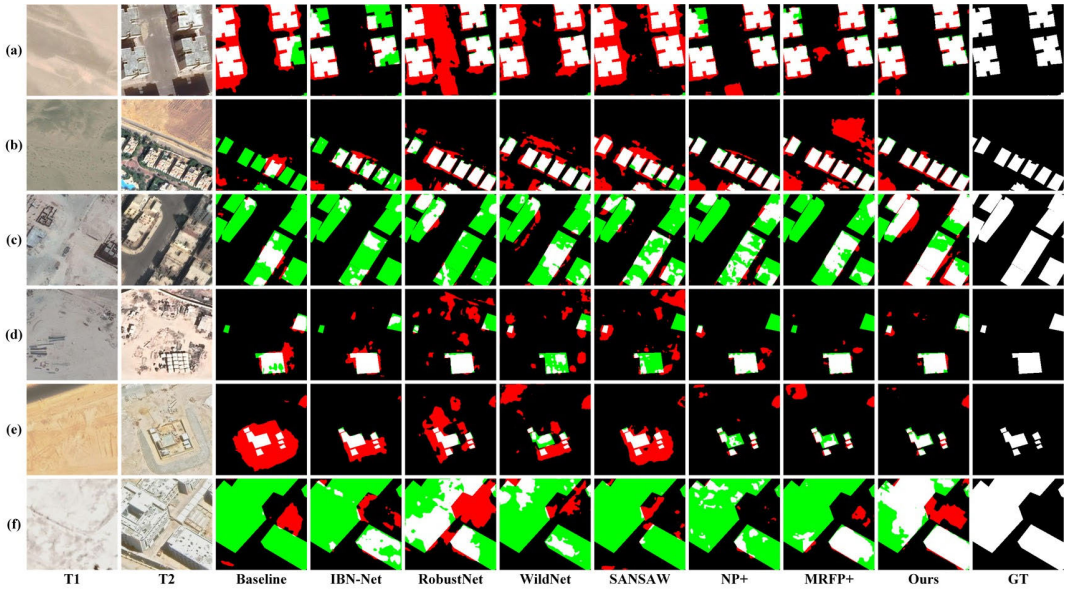


Fig. 8. Qualitative comparison between methods trained on SyntheWorld and tested on unseen target domains of EGY using DeepLabV3+ with ResNet-50 backbone. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (a)–(f) Six representative change detection examples, including the bitemporal images, predictions of different methods, and the corresponding ground truth.

TABLE III

RESULTS OF VARYING THE FEATURE LAYERS FOR APPLYING THE ACML AND CCDL LOSS FUNCTIONS. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

FC Layers	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Baseline	45.61	29.67	40.45	38.86	24.45	34.20	40.00	25.04	35.52
Input Conv(I.C)	51.60	34.48	47.73	44.20	28.75	42.20	43.39	27.25	39.46
ResNet Stage 1(R.S 1)	52.63	35.57	48.91	45.24	29.38	42.15	43.41	27.56	39.40
ResNet Stage 2(R.S 2)	52.19	35.06	48.31	45.52	28.93	42.55	43.43	27.55	38.86
ResNet Stage 3(R.S 3)	49.94	33.16	45.97	42.95	26.25	40.13	42.36	26.05	38.45
I.C+ R.S 1	54.01	37.00	49.42	45.71	29.63	42.67	44.07	28.26	39.12
I.C+ R.S 1+ R.S 2	**54.52**	**37.48**	**49.98**	**46.44**	**30.28**	**43.44**	**44.30**	**28.56**	**39.49**
I.C+ R.S 1+ R.S 2+ R.S 3	40.78	25.72	36.72	42.81	27.25	39.74	42.96	27.41	38.86

capture richer content information. As noted in IBN-Net [9], the shallow layers of convolutional networks (Input Conv, ResNet Stage 1, and ResNet Stage 2) tend to encode style information, and subsequent work such as RobustNet [10] also applies the ISW loss to these shallow layers. In the context of the RSCD, we revisited which layers are most effective for applying the ACML and CCDL losses to extract domain-invariant features. We conducted experiments on the ResNet-50 network using the following layers: Input Conv (output feature size: $64 \times 128 \times 128$), ResNet Stage 1 ($256 \times 64 \times 64$), ResNet Stage 2 ($512 \times 32 \times 32$), and ResNet Stage 3 ($1024 \times 16 \times 16$), with an input image size of $3 \times 256 \times 256$. The results, shown in Table III, indicate that applying ACML and CCDL losses individually to any of the first four convolutional groups improves the model's generalization ability. However, when the ACML and CCDL losses are applied simultaneously to all four groups, the generalization performance decreases. Optimal performance is achieved when the losses are applied to the first three convolutional groups only.

2) Selection of CDL Layers: In the main experiments, we applied CDL to the following feature maps: the final layer feature map of the backbone network (feature size: $2048 \times 16 \times 16$), the feature map output by ASPP (feature size: $304 \times 64 \times 64$), and the feature map obtained by channel-wise concatenation of the upsampled output of the second convolutional group of the backbone network (ResNet Stage 1, MobileNetV2 Bottleneck Layer 1, and ShuffleNetV2 Stage 2) with the ASPP output (feature size: $256 \times 64 \times 64$). Following previous studies, we initialized CDL from the bitemporal fused features obtained by channel-wise concatenation of the Stage 4 outputs from ResNet encoders of the same version at different time points, and then sequentially applied our spatiotemporal consistency contrastive learning loss. Table IV presents the results of applying this method to different network layers. The experiments demonstrate that simultaneously applying the CDL loss to the three intermediate layers yields the best performance.

3) Ablation Study of the Proposed Modules: Impact of different proposed loss functions on generalization performance.

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TABLE IV

RESULTS OF VARYING THE FEATURE LAYERS FOR APPLYING THE STCCL AND PCRCL LOSS FUNCTIONS. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

CDL Layers	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Baseline	45.61	29.67	40.45	38.86	24.45	34.20	40.00	25.04	35.52
ResNet Stage 4(R.S 4)	53.88	36.91	49.41	44.49	28.65	40.36	45.17	29.17	**41.32**
ASPP	54.67	37.68	50.36	46.34	30.20	42.98	45.23	29.31	41.09
Concat	50.95	34.39	46.55	47.52	31.18	44.26	43.87	28.10	39.51
R.S 4+ ASPP	50.27	33.64	46.12	46.47	30.27	43.20	42.58	27.05	38.51
R.S 4+ ASPP+Concat	**55.78**	**38.71**	**51.60**	**47.56**	**31.22**	**44.54**	**45.33**	**29.32**	40.92

TABLE V

ABLATION STUDIES. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

$\mathcal{L}_{ACM}$	$\mathcal{L}_{CCD}$	$\mathcal{L}_{STCCL}$	$\mathcal{L}_{PCRCL}$	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
				F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
				45.61	29.67	40.45	38.86	24.45	34.20	40.00	25.04	35.52
√				52.20	35.37	48.37	45.93	29.84	42.43	43.02	27.44	39.11
√	√			54.52	37.48	49.98	46.44	30.28	43.44	44.30	28.56	39.49
		√		55.28	38.26	50.79	47.18	30.87	43.84	44.33	28.53	40.14
		√	√	55.78	38.71	51.60	47.56	31.22	44.54	45.33	29.32	40.92
√	√	√		55.48	38.41	51.84	48.02	31.60	44.96	44.41	28.55	40.67
√	√	√	√	**56.33**	**39.25**	**52.37**	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**

TABLE VI

CALCULATION OF CCDL. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Cross-covariance diagonal loss								
	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Whitening	51.20	34.42	47.34	49.29	32.73	46.42	43.92	28.19	40.26
Diagonal	**56.33**	**39.25**	**52.37**	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**

We trained the baseline using ResNet-50 as the backbone on the SynthWorld dataset and evaluated it on the GZ, WHU, and EGY datasets, with the results presented in Table V. Specifically, the baseline performed poorly on the target domains due to overfitting to the feature distribution of the synthetic data. In contrast, integrating any of the proposed loss mechanisms led to significant performance improvements. More specifically, the introduction of FC ($\mathcal{L}_{ACM}, \mathcal{L}_{CCD}$), which associates features of paired images from the same time point, demonstrated its effectiveness in preserving critical information. The effects of CDL ($\mathcal{L}_{STCCL}, \mathcal{L}_{PCRCL}$) were also notable, as it constructs a more discriminative embedding space, facilitating the correction of features associated with pseudo changes.

a) Auto-covariance matching loss: Fig. 9 illustrates the t-SNE [48] plots of auto-covariance for the original and augmented images across two temporal phases, before and after applying ACML. The baseline exhibits perceptual differences between the original and augmented images of the same temporal phase from a style perspective. However, after applying ACML, the feature distributions of the original and augmented images become more closely aligned, indicating that the proposed ACML effectively preserves content recognition.

b) Cross-covariance diagonal loss: Table VI shows that our proposed CCDL method achieves better performance by converging the diagonal elements to 1. As mentioned earlier, removing off-diagonal elements that contain content information can actually lead to performance degradation.

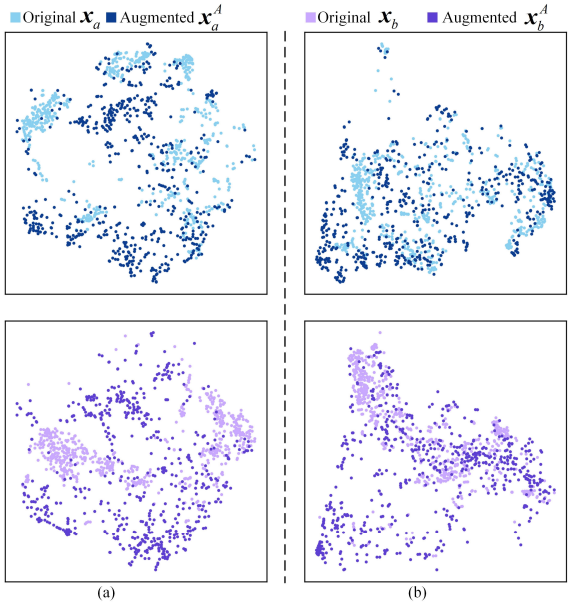


Fig. 9. t-SNE [48] visualization comparing the auto-covariance with and without ACML. (a) Baseline. (b) Ours.

c) Sampling number in STCCL: Tables VII and VIII illustrate the effects of the number of sampled classes per image and the number of samples per class on performance in STCCL, respectively. As shown in Table VII, performance improves with increased diversity of sampled classes, indicating that contrasting across a greater number of classes

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TABLE VII

NUMBER OF CLASSES FOR STCCL. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

#	# of classes								
	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
1(unchange)	53.25	36.30	49.33	43.02	32.02	45.57	44.79	28.88	41.12
1(change)	54.76	37.71	50.63	46.04	29.91	42.75	47.35	31.02	43.14
2	**56.33**	**39.25**	**52.37**	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**

TABLE VIII

NUMBER OF NEGATIVE SAMPLES FOR STCCL. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

#	# of negative samples								
	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
10	**56.33**	**39.25**	**52.37**	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**
50	53.75	36.83	49.70	47.54	31.23	44.29	45.02	29.09	40.91
100	47.64	31.29	43.56	47.96	31.57	45.10	43.01	27.45	39.26

TABLE IX

PROJECTION HEAD OF PCRCL. "SG" INDICATES STOP GRADIENT. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

MLP	Projection Head of PCRCL								
	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Individual	49.19	32.64	44.82	45.93	29.83	42.85	48.35	31.92	44.52
Shared(SG)	53.82	36.82	49.66	46.89	30.63	43.64	46.29	30.24	42.34
Shared	**56.33**	**39.25**	**52.37**	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**

TABLE X

QUANTITATIVE ANALYSIS OF FEATURE CONSISTENCY AND CLASS DISCRIMINABILITY. RESULTS ARE AVERAGED OVER THREE FEATURE LEVELS AND THREE RANDOM TRIALS USING 2000 SAMPLES PER CLASS. LOWER MMD^2 AND HIGHER FDR ARE BETTER

Methods	MMD^2 ($\times 10^{-4}$) ↓	FDR ↑
Baseline	5.0805	0.8262
Baseline +FC	1.8296	-
Baseline +CDL	-	1.0617

enhances the model's discriminative capability. Furthermore, Table VIII shows that an excessive number of negative samples per class can adversely affect model performance.

d) Projection head in PCRCL: We investigated the impact of different projection head configurations on PCRCL. Three setups were tested: an independent projection head, a duplicated STCCL projection head with gradient stopping, and a projection head shared with STCCL. As shown in Table IX, sharing the projection head achieved the best results. The results indicate that PCRCL not only relies on the semantic information provided by STCCL to effectively correct pseudo-change features but also enhances the embedding space established by STCCL.

e) FC module: As shown in Table X, FC reduces the squared maximum mean discrepancy (MMD^2) by approximately 64.0%. This result indicates that FC effectively mitigates augmentation-induced distribution shifts and improves feature consistency across the two branches.

f) CDL module: Table X also shows that CDL increases the Fisher discriminant ratio (FDR) by approximately 28.5%. The higher FDR demonstrates that CDL enhances interclass separability relative to intraclass dispersion, thereby producing more discriminative representations for changed and unchanged regions.

F. Analysis of Loss Function Coefficients

We adjusted the weight parameters of each loss function in (14) and conducted experiments with a step size of 0.1, using the initially best-performing model configuration as the baseline. The results are presented in Table XI. Each loss function is highly sensitive to its corresponding coefficient. Specifically, $\mathcal{L}_{ACM}$ and $\mathcal{L}_{CCD}$, as style-invariant components, need to be maintained at appropriate levels to preserve useful structural information, whereas $\mathcal{L}_{STCCL}$ and $\mathcal{L}_{PCRCL}$ exhibit better performance when moderately strengthened. Using the $\omega_1 = 0.2$, $\omega_2 = 0.2$, $\omega_3 = 0.4$, and $\omega_4 = 0.3$ configuration yielded optimal performance for the SyntheWorld → GZ scenario, while the $\omega_1 = 0.2$, $\omega_2 = 0.2$, $\omega_3 = 0.3$, and $\omega_4 = 0.3$ configuration provided the best results for the SyntheWorld → EGY and SyntheWorld → WHU scenarios. Considering the overall performance across all datasets, we ultimately selected $\omega_1 = 0.2$, $\omega_2 = 0.2$, $\omega_3 = 0.3$, and $\omega_4 = 0.3$.

G. Cross-Domain Scenario Applicability Analysis

In this section, we further validate the applicability of the proposed mechanism across various cross-domain scenarios. As shown in Tables XII–XIV, we adopt the DeepLabV3+ network with a ResNet-50 backbone as the baseline, using three real-world datasets individually as source domains while designating the remaining three datasets as target domains, thereby constructing nine cross-domain combinations. The experimental results demonstrate that our method consistently

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TABLE XI
SENSITIVITY TO WEIGHTING PARAMETERS OF EACH LOSS FUNCTION. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

$\omega_1(\mathcal{L}_{ACM})$	$\omega_2(\mathcal{L}_{CCD})$	$\omega_3(\mathcal{L}_{STCCL})$	$\omega_4(\mathcal{L}_{PCRCL})$	SyntheWorld → GZ			SyntheWorld → WHU			SyntheWorld → EGY		
				F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
0.2	0.2	0.3	0.3	56.33	39.25	52.37	**50.13**	**33.48**	**47.07**	**48.73**	**32.28**	**44.86**
0.1	0.2	0.3	0.3	52.56	35.67	47.98	45.54	29.50	42.08	41.06	25.99	36.36
0.3	0.2	0.3	0.3	50.21	33.54	45.81	44.94	29.01	41.53	44.13	28.38	39.58
0.2	**0.1**	0.3	0.3	49.32	32.74	45.36	47.50	31.15	44.53	42.95	27.39	39.09
0.2	**0.3**	0.3	0.3	49.13	32.78	45.19	44.94	29.05	41.90	44.45	28.68	40.67
0.2	0.2	**0.2**	0.3	49.61	33.03	45.24	46.78	30.59	43.53	39.07	24.36	34.71
0.2	0.2	**0.4**	0.3	**56.95**	**39.83**	**52.80**	46.34	30.17	42.80	47.40	31.08	43.19
0.2	0.2	0.3	**0.2**	51.64	34.89	47.14	46.15	30.02	42.68	44.78	28.87	40.11
0.2	0.2	0.3	**0.4**	47.34	31.07	42.90	45.88	29.78	42.70	44.86	28.95	40.79

TABLE XII
TEST RESULTS AFTER TRAINING ON THE GZ

Methods	GZ → EGY			GZ → WHU			GZ → SyntheWorld			GZ → GZ		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Baseline	46.42	30.23	43.37	38.58	23.90	34.44	20.35	11.33	17.87	82.81	70.66	81.26
Ours	**48.11**	**31.67**	**44.13**	**50.31**	**33.61**	**46.98**	**26.26**	**15.11**	**22.75**	**83.01**	**70.95**	**81.35**

TABLE XIII
TEST RESULTS AFTER TRAINING ON THE WHU

Methods	WHU → EGY			WHU → GZ			WHU → SyntheWorld			WHU → WHU		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Baseline	7.93	4.13	5.34	58.16	41.01	55.10	9.69	5.09	8.78	81.15	68.28	80.19
Ours	**26.44**	**15.23**	**22.49**	**58.69**	**41.54**	**55.37**	**16.96**	**9.27**	**13.39**	**82.63**	**70.40**	**81.68**

TABLE XIV
TEST RESULTS AFTER TRAINING ON THE EGY

Methods	EGY → WHU			EGY → GZ			EGY → SyntheWorld			EGY → EGY		
	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Baseline	46.61	30.39	44.54	40.00	25.00	36.95	35.82	21.82	33.19	75.21	56.51	70.47
Ours	**47.84**	**31.44**	**45.42**	**55.73**	**38.63**	**52.61**	**43.89**	**28.12**	**39.93**	**76.58**	**62.05**	**74.81**

TABLE XV
COMPUTATIONAL COST COMPARISON CONDUCTED USING DEEPLABV3+ WITH A RESNET-50 BACKBONE ON AN NVIDIA RTX 4090 GPU WITH AN IMAGE RESOLUTION OF 256 × 256. INFERENCE TIME IS AVERAGED OVER 400 TRIALS

Methods	External		Params(M)	GFLOPS	Time(ms)
	Dataset	Module			
Baseline	-	-	117.03	49.71	23.53
IBN-Net	-	-	192.40	49.72	21.13
RobustNet	-	-	192.40	49.72	21.24
WildNet	√	-	117.03	49.71	23.53
SANSAW	-	√	122.80	70.15	39.61
NP+	-	-	117.03	49.71	23.53
MRFP+	-	-	117.03	49.71	23.53
Ours	-	-	117.03	49.71	23.68

and significantly improves the DG performance of the baseline across all settings, while simultaneously enhancing its performance on the source domain. These findings fully verify the robustness and universality of the proposed mechanism in complex cross-domain tasks.

H. Computational Cost and Efficiency Analysis

As shown in Table XV, we compared the number of parameters and computational cost of the proposed method with existing DG approaches. Since our method introduces only the augmented bi-temporal image pairs and four loss functions during training, and fully relies on the original baseline RSCD model DeepLabV3+ during inference, its number of parameters (Params) is identical to that of the baseline model. Similarly, the computational cost (GFLOPS) remains consistent with the baseline, without introducing any additional inference overhead, facilitating practical deployment. In summary, our approach intrinsically learns features without requiring separate modules, minimizing dependency on inference-stage resources and providing superior scalability and deployment advantages in practical applications.

V. DISCUSSION

A. Advantages Compared to Other DG Methods

In addition, compared with existing DG methods, our approach offers several advantages. First, it effectively addresses the common issue of content information loss in FN methods, thereby extracting style-invariant features while preserving semantic integrity. Second, it mitigates the degradation of discriminative capability caused by domain shifts, thereby enhancing model robustness. Third, it exhibits stability across CNNs with varying computational resources and time constraints, making it adaptable to diverse scenarios in terms

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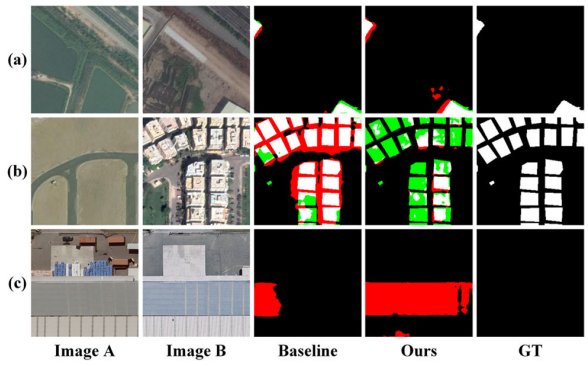


Fig. 10. Analysis of selected failure cases. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (a) Demonstrates model predictions on GZ, (b) demonstrates model predictions on EGY, and (c) demonstrates model predictions on WHU datasets.

of computational efficiency and real-time requirements. Moreover, the method requires no auxiliary domains or external modules, significantly reducing inference-time resource and time overhead.

B. Limitations and Future Work

Although the proposed CDCCNet demonstrates strong performance in RSCD, there are still some limitations that can be addressed in future work as follows.

1) In Fig. 10(a), more false positives occur around target shadows, suggesting confusion between illumination-induced radiometric variations and genuine changes. In Fig. 10(b), extensive omissions are observed for densely distributed buildings, indicating a limited ability to distinguish adjacent targets. In Fig. 10(c), interior roof regions are falsely detected because pronounced intraobject color and texture variations may be mistaken for temporal changes. Future work will focus on improving robustness to shadows and material variations and developing cross-domain consistency learning that preserves high-resolution details and multiscale semantics, thereby reducing pseudo-change false positives and missed detections in unseen domains.

2) Since no multiple datasets of other land-cover categories for RSCD could be found that are different yet related (i.e., sharing the same feature and label spaces but exhibiting different probability distributions), it was not feasible to establish an experimental setting for DG change detection. Therefore, the experiments in this article only demonstrate the performance and applicability of the proposed method in building change detection. However, based on the fundamental principles of the proposed model, its applicability is not limited to building change detection but can also be extended to other land-cover categories. In future research, we will gradually present applications of our approach to RSCD involving other land-cover categories. In addition, the model is sensitive to loss weighting coefficients, and future research could focus on improving and stabilizing the FC approach.

VI. Conclusion

In this article, we propose a new DG mechanism for RSCD, which aims to transfer a change detection model trained on a fully annotated dataset (source domain) to any unseen dataset (target domain) without relying on target domain samples or labels. Extensive experimental results demonstrate that, regardless of whether the training is conducted on synthetic or real datasets, the proposed mechanism consistently outperforms existing state-of-the-art methods across various cross-domain scenarios. Moreover, when employing three different backbones with varying computational resource requirements, our method achieves either the best or the second-best performance, highlighting its robust superiority and broad applicability. Besides these, the proposed approach not only enhances DG performance on arbitrary unseen target domains but also improves testing performance within the source domain.

Acknowledgment

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